



CONTACT

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GitHub
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EXPERTISE

Full Stack Development

- React, TypeScript
- Django, Node.js
- MySQL, Redis

DevOps Engineering

- Docker, Kubernetes
- Helm, Harbor, Nginx
- GitHub Actions, CI/CD

AI & Data Engineering

- AI Agent, Tool Use
- N8N, Dify
- Kafka, KubeFlow

PERSONALITY

Detail-oriented:

Focused on robustness and comprehensiveness when architecting system orchestration and scheduling workflows.

Self-motivated:

Passionate about deep-diving into complex technical bottlenecks with long-term research capability.

HUANG, YAO-CHING (Kim)

Master's Student / Full Stack & AI Engineer

ABOUT MYSELF

Experienced in full-stack development, DevOps deployment, and AI Agent development. Explored RL to enhance SSD performance, integrated Dify with N8N for automated workflows, and operated Kubernetes clusters for 6G platform. Skilled in containerized deployment, CI/CD pipelines, and cross-platform system integration. Self-motivated and dedicated to applying academic knowledge to practical development.

EXPERIENCE

SparkAmplify — Winter Intern (US) Work

Built frontend-backend separated deployment with Docker, Nginx reverse proxy, and Harbor image registry. Established Git Workflow and CI/CD pipeline with GitHub Actions. Authored platform architecture docs and deployment diagrams.

6G AI/ML Platform — O-RAN x KubeFlow x Agent AI Project

Operated 20+ container services across dual-node hybrid deployment (Docker + Kubernetes). Demonstrated at EuCNC 2024 (Belgium) and COSCUP 2024. Integrated intent-driven platform with AI Agent to control the system.

MySphere — English Story Learning Platform Project

Built learning progress tracking API and LLM-powered Q&A Chat. Set up Docker + Nginx deployment, Harbor image management, and GitHub Actions CI/CD for the team.

Calculus TA — Grade & Exam System Work

Independently built a grade registration & exam management system (React + Django + DB). Developed auto-exam AI Agent with Tool Use, embedding Word tool to autonomously generate exam documents.

EDUCATION

National Taiwan University of Science and Technology (NTUST)

Sep 2025 - Present

M.S. in Electronic Engineering

National Taiwan University of Science and Technology (NTUST)

Sep 2021 - Jun 2025

B.S. in Electronic Engineering

- GPA: 89.73 / 100 | Class Rank: 5/37 (Top 13.5%) | Dept. Rank: 13/109 (Top 11.9%)

RESEARCH & THESIS

Improving SSD GC Long Tail Latency Using Idle Time B.S. Thesis

Applied Deep Q-Learning for RL-based GC scheduling; validated with 100K SNIA IOTTA traces on MQSim. Bridged Python/C++ via Cython; effectively reduced long tail latency occurrence.

O-RAN SC Dual-Layer Architecture Deployment Research

Deployed SMO/OAM layer (3 Docker Compose Stacks) and NonRT RIC layer (13 Helm subcharts on K8s). Implemented Docker x Kubernetes cross-platform network integration.

HONORS

Study Abroad Dream Program — SparkAmplify Winter Internship (2026)
Space & ICT New Star Program — Industrial Development Admin, MOEA (2024)